

ORTIS YANKEY, Ph.D.

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Professional Summary

I am a geospatial data scientist and population health researcher with over five years of post PhD experience in the field of spatial epidemiology, population modelling, environmental health analysis, and public health analysis. My work sits at the intersection of data science, public health, and spatial analytics, leveraging machine learning, Bayesian statistics, advanced generalized linear models, and geospatial methods to tackle population health research questions for policy decisions. I have gained experience in engaging stakeholders and enhancing their capacity in both national and international projects.

Key Skills

- **Software & Analytics:** Advanced proficiency in using R, SAS, and SPSS for data analysis.
- **Domain Expertise:** Machine learning, Advanced statistics, Bayesian statistics, Spatial statistics
- **Machine learning:** Supervised and unsupervised classification, ensemble learning, and artificial neural networks
- **Advanced Statistics:** Regression models, Bayesian modelling, model validation, and uncertainty quantification.
- **Geospatial Analytics:** ArcGIS Pro, QGIS, Geoda, ENVI, Remote Sensing, Spatial Statistics
- **Earth Observation:** Image processing and classification, image indices estimation
- **IT Skills:** Workflow deployment in high-performance computers, knowledge of GitHub
- **Project Leadership & Collaboration:** Interdisciplinary research, stakeholder engagement, technical training

Education

- PhD Geography (Spatial Statistics), Kent State University, USA - May 2022
- MSc Geographic Information Systems (GIS), University of Akron, USA – May 2018
- MSc Urbanisation & Development, London School of Economics, UK – November 2015
- BSc Land Economy, KNUST, Ghana – May 2013

Professional Experience

Senior Research Fellow

June 2026 - Current

WorldPop Research Group, University of Southampton, UK

Responsibilities

- Develops advanced statistical models for predicting health outcomes at smaller geographic scales.
- Lead and co-develop advanced machine learning and Bayesian statistical models to estimate population and demographic indicators at small area scales.
- Direct the integration of multi-source data streams, including census, survey, and earth observation data, to pioneer novel spatial analysis techniques for health and population outcome analysis
- Mentor and guide junior researchers, PhD candidates, and data analysts within WorldPop to

foster professional growth

- Lead training development for national statistical offices on data science workflows, health outcome modelling and population modelling using R and GIS.

Research Fellow

January 2022 – June 2026

WorldPop Research Group, University of Southampton, UK

Responsibilities

- Led and co-developed machine learning and Bayesian statistical models to estimate population and demographic indicators at small area scales.
- Developed advanced statistical models for predicting health outcomes at smaller geographic scales.
- Conducted simulation studies to ascertain model validity and replicability
- Integrated census, survey, and earth observation data for spatial analysis and predicting health outcomes.
- Delivered training to national statistical offices on data science workflows and population modelling using R and GIS.

Graduate Teaching Assistant/Ph. D. Researcher

August 2018 – January 2022

Kent State University, OH USA, Department of Geography

Responsibilities

- Designed and implemented Bayesian spatial models for analyzing urban and public health datasets.
- Conducted literature reviews, questionnaire design, and manuscript preparation.
- Taught undergraduate students' various courses in GIS/Geography
- Organized lab work and tutorial sessions for these classes.
- Graded submitted test materials.
- Held office hours for students.

Graduate Teaching Assistant

August 2016 – May 2018

The University of Akron, OH USA, Department of Geosciences

Responsibilities

- Assisted students with their GIS tutorials and lab assignments.
- Proctored exams and assisted with the grading of tests.
- Helped students with their data analysis using statistical software such as R and SPSS

Selected Project in WorldPop

In my role as a Researcher at the WorldPop Research Group (University of Southampton), I have led several projects as a lead or co-lead modeller in several projects. Below I present some of the recent projects I have been engaged in.

- **Health and Demographic Surveillance System Population Forecast for Malaria Vaccination Campaigns - Lead Modeller**
- Leading a machine learning and bespoke Bayesian approach to forecast population size and demographics in HDSS catchment districts in Ghana (*2026 output planned*)

- **Count-Down to 2030 – Geospatial Modelling of Maternal Health Outcomes Across Sub Saharan Africa - *Lead Modeller***
Led geostatistical modelling of antenatal care visits among pregnant women in Sub-Saharan Africa (2025)
- **Geospatial Modelling of Vaccine Coverage in Nigeria and Ethiopia – *Co-lead Modeller***
Applied Bayesian and machine learning approaches to compare estimates of vaccine coverages in Nigeria and Ethiopia. (2025)
- **Estimation of Populations Targeted for Vaccination in Cameroon—*Lead Modeller***
Led geospatial modelling to estimate children targeted for vaccination campaigns using microcensus and geospatial data. (2025 output planned)
- **Gridded Population Modelling for DRC Vaccination Campaigns – *Co-lead Modeller***
Coordinated data integration and spatial modelling of population count to support national

Teaching Experience

Teaching Summary

My previous and current teaching activities include leading courses, developing new courses, designing teaching materials, delivering lectures, developing GIS practical guidelines, leading practical sessions, designing assessments and grading students' assignment, providing feedback, course evaluation and coordinating teaching and learning activities.

Current Teaching (2025 – 2026)

- Core Skills in GIS (Fall 2026) – Masters, University of Southampton
- Core Skills in GIS (Fall 2025) – Masters, University of Southampton

Previous Teaching (2017 – 2021)

- World Geography (Fall 2021) - Undergraduate, Kent State University
- Introduction to Geography (Spring, 2021) - Undergraduate, Kent State University
- Introduction to GIS (Fall 2020) - Undergraduate, Kent State University
- Physical Geography Lab (Spring 2020) - Undergraduate, Kent State University
- Spatial Analysis (Spring 2018) – Undergraduate, University of Akron (Teaching Assistant)
- Introduction to GIS (Fall 2017) – Undergraduate, University of Akron (Teaching Assistant)

Training and Workshops Conducted

- Workshop on Generating Socio-Demographic Data in South-Central Somalia – Somali National Bureau of Statistics, March 2026 (**Lead Instructor**)
- Introduction to R for Statistical Analysis – Postgraduate Students, University of Southampton, January 2026, May 2025 and February 2024 (**Lead Instructor**)
- Workshop/Capacity Strengthening and Co-Development of Modelled Population Estimates for Hybrid Census Implementation in Cameroon using Field Test Data, December 2025 (**Co – Instructor**)
- Adjusting for Census Undercount in Jamaica – Statistical Institute of Jamaica March 2025 (**Lead Instructor**)

- Gridded Population Modelling Workshop – Cameroon National Statistical Office (NSO) April 2024 (**Co-Instructor**)
- Spatial Population Modelling Training – Thailand National Statistical Office November 2023 (**Co-Instructor**)
- Modelled Population Estimates and National Sampling Frames – Training for census teams and NSOs across Latin America and the Caribbean June 2023 (**Lead Instructor**)
- Capacity-Strengthening Workshop on Population Modelling – Nigeria Population Commission February 2023 (**Co-Instructor**)

Research Interest

- Spatial Epidemiology (Health Geography) for Targeted Interventions
- Spatiotemporal Models for Disease Surveillance
- Spatial Demography (Population Modelling) for Resource Allocation
- Applied Statistics in Health Data Analysis
- Data Analytics for Health Systems

Publications

1. Dotse-Gborgbortsi, W., **Yankey, O.**, Utazi, E. C., Tejedor-Garavito, N., Banke-Thomas, A., Chaudhuri, S., ... & Nilsen, K. (2026). Mapping zero maternal and newborn health service utilisation across 34 African countries: a geostatistical analysis of household survey data. *VeriXiv*, 3(103), 103.
2. Gadiaga, A. N., Tine, M. W., Diene, A. N., Linard, C., Speybroeck, N., **Yankey, O.**, ... & Tatem, A. J. (2026). Spatio-temporal modelling of COVID-19 infection and associated risk factors in Dakar, Senegal. *PLOS Global Public Health*, 6(6), e0004945.
3. Tatem, A. J., Boo, G., Chamberlain, H. R., Nnanatu, C. C., Darin, **E., Yankey, O.**, ... & Lázár, A. N. (2026). Advances in small area population estimation in the absence of national census data. *Proceedings of the National Academy of Sciences*, 123(35), e2413993123.
4. Pompili, L., Sorichetta, A., Economou, T., Bondarenko, M., **Yankey, O.**, & Camera, C. A. (2026). Evaluating the use of a coordinate-based tensor-product for addressing spatial autocorrelation in shallow landslide susceptibility modelling. *Stochastic Environmental Research and Risk Assessment*, 40(6), 157.
5. **Yankey, O.**, Essah, M., & Amegbor, P. M. (2025). The COVID-19 pandemic and self-reported food insecurity among women in Burkina Faso: evidence from the performance monitoring for action (PMA) COVID-19 survey data. *BMC Women's Health*, 25(1), 42.
6. Utazi, C. E., **Yankey, O.**, Chaudhuri, S., Olowe, I. D., Danovaro-Holliday, M. C., Lazar, A. N., & Tatem, A. J. (2025). Geostatistical and machine learning approaches for high-resolution mapping of vaccination coverage. *Spatial and Spatio-temporal Epidemiology*, 100744. <https://doi.org/10.1016/j.sste.2025.100744>.
7. Dotse-Gborgbortsi, W., Nilsen, K., **Yankey, O.**, Ofosu, A., Ankomah, T., Tweneboah, E., ... & Wright, J. (2025). Spatio-temporal patterns of health service delivery and access to maternal, child, and outpatient healthcare in Volta region, Ghana: a repeated cross-sectional ecological study using health facility data. *Global Health Action*, 18(1), 2513861.

8. Nnanatu, C. C., Bonnie, A., Joseph, J., **Yankey, O.**, Cihan, D., Gadiaga, A., ... & Tatem, A. J. (2025). Estimating small area population from health intervention campaign surveys and partially observed settlement data. *Nature Communications*, 16(1), 1-13.
9. **Yankey, O.**, Utazi, C. E., Nnanatu, C. C., Gadiaga, A. N., Abbot, T., Lazar, A. N., & Tatem, A. J. (2024). Disaggregating Census Data for Population Mapping Using a Bayesian Additive Regression Tree Model. *Applied Geography*, Volume 172(103416), <https://doi.org/10.1016/j.apgeog.2024.103416>.
10. Buckman, T. A., Sakyi, S. A., Yeboah-Mensah, K., Antwi, M. H., Darban, I., Owusu-Brenya, L., & **Yankey O.** (2024). Demographic, Clinical Profile of Rheumatoid Arthritis Patients and Their Association with Disease Severity in Ghana. *International Journal of Rheumatology*, 2024.
11. **Yankey, O.**, Lee, J., Gardenhire, R. & Borawski E (2023). Neighborhood Racial Segregation Predict the Spatial Distribution of Supermarkets and Grocery Stores Better than Socioeconomic Factors in Cleveland, Ohio: a Bayesian Spatial Approach. *J. Racial and Ethnic Health Disparities*. <https://doi.org/10.1007/s40615-023-01669-4>
12. Botchwey, G., & **Yankey, O.** (2023). Small-Scale Mining, Environment and Livelihoods: Perspectives from Mining Communities in Ghana. *African Journal of Social Sciences Education*, 2(2).
13. **Yankey, O.**, Amegbor, P. M., & Lee J. (2021). The Effect of Sociodemographic Factors on the Risk of Poor Mental Health in Akron (Ohio): A Bayesian Hierarchical Spatial Analysis. *Journal of Spatial & Spatiotemporal Epidemiology*, 100438. <https://doi.org/10.1016/j.sste.2021.100438>
14. **Yankey O.**, Amegbor, P. M., & Essah M. (2021). The Effect of Socioeconomic and Environmental Factors on Obesity: A Spatial Regression Analysis. *International Journal of Applied Geospatial Research (IJAGR)*
15. Amegbor, P.M., **Yankey, O.**, Davies, M. & Sabel, C. E. (2021). Individual and contextual predictors of overweight or obesity among women in Uganda: a spatio-temporal perspective. *Geojournal*. <https://doi.org/10.1007/s10708-021-10466-7>
16. Amegbor, P. M., **Yankey, O.**, & Sabel, C. E. (2020). Examining the effect of geographic region of residence on childhood malnutrition in Uganda. *Journal of tropical pediatrics*, 66(6), 598-611.
17. Amegbor, P. M., **Yankey, O.**, Rosenberg, M. W., & Sabel, C. E. (2020). Examining spatial variability in the association between male partner alcohol misuse and intimate partner violence against women in Ghana: a GWR analysis. *Journal of interpersonal violence*, 0886260519900299.

Book Chapter

18. **Yankey, O.**, Hu T, Yue H, Wang P, Xu X. (2023). Spatiotemporal Bayesian Regression. A book chapter in *Spatiotemporal Analytics*. CRC Press, Taylor & Francis Group
19. WorldPop. 2024. WorldPop Book of Methods, Vol. I: Gridded Population Estimates. WorldPop, University of Southampton. 08 October 2024 (<https://wpgp.github.io/bookworm/>)

Manuscript Under Review/Pre-Prints

- Brzezinska I., Davila P., Jasper P., Lindsay N., **Yankey O.** (2026). High Spatial Resolution Mapping of Early Childhood Development Indicators.

- Adewole, W. A., **Yankey, O.**, Utazi, E., Lloyd, C., Cockings, S., & Tatem, A. J. (2026). Classifying urban areas into residential, non-residential and mixed-use proportions using building footprints and geospatial models.
- Nnanatu, C. C., **Yankey, O.**, Dzossa, A. D., Abbott, T., Gadiaga, A., Lazar, A., & Tatem, A. (2025). Efficient Bayesian hierarchical small area population estimation using INLA-SPDE: integrating multiple data sources and spatial-autocorrelation.
- Nnanatu, C. C., **Yankey, O.**, Dzossa, A. D., Abbott, T., Gadiaga, A., Lazar, A. N., & Tatem, A. J. (2025). Census-independent small area estimates of population and number of households in Cameroon. *VeriXiv*, 2(88), 88.

Conference Presentations

1. **Yankey, O.** (presenting author), Nnanatu, C. C., Gadiaga, T., Lazar, A. N., & Tatem, A. J. (2025). Innovative Approaches to Population Modelling: A Case Study of the DRC. American Association of Geographers Annual Conference, Detroit, 2025
2. **Yankey, O.** (presenting author), Utazi, C. E., Nnanatu, C. C., Gadiaga, A. N., Abbot, T., Lazar, A. N., & Tatem, A. J. (2024). Disaggregating Census Data for Population Mapping Using a Bayesian Additive Regression Tree Model. GISRUK Conference, Leeds, 2024
3. **Yankey, O.** (presenting author), Amegbor, P. M., & Lee J. (2021). The Effect of Sociodemographic Factors on the Risk of Poor Mental Health in Akron (Ohio): A Bayesian Hierarchical Spatial Analysis. Healthy Communities Research Initiative, Kent State University, March 8-9, 2021. *Moved to virtual due to COVID 19
4. Amegbor, P. M., **Yankey, O.** (presenting author), & Sabel, C. E. (2020). Examining the effect of geographic region of residence on childhood malnutrition in Uganda. Association of American Geographers Annual Meeting, Denver (Colorado) April 4-10, 2020. *Moved to virtual due to COVID 19
5. Amegbor, P. M., **Yankey, O.** (presenting author), Rosenberg, M. W., & Sabel, C. E. (2020). Examining spatial variability in the association between male partner alcohol misuse and intimate partner violence against women in Ghana: a GWR analysis. Association of American Geographers Annual Meeting, Washington-D.C, April 3-7, 2019
6. Yankey O.(Poster Presentation) Using Geostatistics to Predict Soil Lead Distribution in Akron (Ohio), and Implications for Urban Gardens. Association of American Geographers Annual Meeting, New Orleans (Louisiana), April 9-14, 2018

Recent Invited Talks

- Using Geospatial Data to Make Global Health Policy Decisions, February 2026 – Invited Seminar, Department of Public Health Sciences (Queen’s University, Canada),
- High Spatial Resolution Mapping of Early Childhood Development Indicators, January 2026 – Invited Seminar, Oxford Policy Management
- Geospatial Data Integration to Produce Gridded Population Estimates at Small Area Scales, April 2025 – Invited Seminar, Department of Geography (Kent State University, USA)

Grants

Dates	Role	Title	Funding Body	Value
July, 2026	Principal Investigator	HDSS Catchment Area Population	Gates Foundation	£55,000

		Forecasting for Malaria Surveillance in Ghana		
October, 2025	Co-Investigator	Technical Assistance & Training on Spatial Statistics and Hybrid Census in Cameroon	United Nations Population Fund (UNFPA)	£107,663.58
August, 2025	Co-Investigator	Technical Assistance for Automated EA Census Mapping in Malawi	Foreign Commonwealth Development Office (FCDO)	£123,000
July, 2025	Co-Investigator	Geospatial Population Estimation for the Caribbean	Foreign Commonwealth Development Office (FCDO)	£350,000

Awards and Recognition

- Commonwealth Shared Scholarship (LSE, UK)
- Graduate Assistantships (Kent State and University of Akron, USA)
- African Community Scholarship for Academic Excellence (Kent State, 2019)

Affiliations

- The Association for Geographic Information, UK
- Commonwealth Scholars Network, UK

Reviewer

- Journal of Racial and Ethnic Health Disparities
- PLOS ONE
- Journal of Health, Population and Nutrition

References

References will be provided upon request